

Yaswanth Muppalla

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Senior Software Engineer | Java/Python Backend | APIs & Distributed Systems | AWS | AI-Enabled Platforms

PROFESSIONAL EXPERIENCE

Senior Software Engineer | American Express

Apr 2023 - Present

Impact: Built Java/Python backend platform services and internal APIs processing 10M+ financial records, improved reporting efficiency by 35%, achieved 99% data accuracy, and reduced rerun/debug cycles by 3x.

- Built Java/Python backend services and internal APIs over SQL and MemSQL to consolidate 10M+ financial records across reporting and reconciliation workflows.
- Designed JSON-configurable service components for ingestion, validation, transformation, and reconciliation, reducing repeated backend code changes.
- Implemented Apache Flink parallel processing and batch/stream patterns for high-volume financial workloads with 99% data accuracy.
- Automated Apache Airflow orchestration with restartable jobs, dependency controls, failure handling, and runbooks, cutting rerun/debug cycles by 3x.
- Improved backend reliability with validation, logging, SLA checks, lineage, auditability, and production triage controls across financial workflows.
- Built internal Python AI copilots over pipeline metadata, run states, quality checks, and runbooks to assist support triage and root-cause analysis.

Software Engineer | Skechers

Oct 2020 - Mar 2023

Impact: Delivered 15+ Python APIs, maintained 99.9% uptime, reduced infrastructure costs by 22%, migrated 35M+ records, and cut incident resolution time by 40%.

- Developed 15+ production REST APIs with Python and Pydantic on AWS Lambda and Step Functions for loyalty/campaign workflows, maintaining 99.9% uptime.
- Applied layered service architecture across serverless applications, separating validation, orchestration, data access, retry handling, and integrations.
- Migrated 35M+ production records from SFMC to MongoDB, improving query performance by 50% through optimized indexing and caching strategies.
- Architected a loyalty rewards platform using SessionM APIs to process daily transactions under 500ms latency, contributing to a 20% increase in customer retention.
- Implemented real-time monitoring with CloudWatch, Datadog, alerts, and API failure tracing, reducing incident resolution time by 40% and maintaining SLA compliance.
- Integrated AWS Step Functions workflows to coordinate async processing, retries, and downstream service calls across loyalty and marketing data operations.

Full-Stack Developer | SOCH Inc. - Richmond, VA

Jan 2020 - Sep 2020

- Developed a Spring Boot-based ticketing system that streamlined issue intake, tracking, and resolution workflows for government clients.
- Built Spring REST microservices with API controllers, validation, exception handling, and service-layer separation to improve scalability.
- Used Java concurrency patterns with ExecutorService/CompletableFuture for async ticket notifications, background jobs, and thread-safe workflow updates.
- Built full-stack workflow features across backend services and issue management screens to improve usability for operational teams.

FOUNDER-LED TECHNICAL PROJECTS

OmInsights - Founder-Led Backend Platform Build

Jan 2024 - Present

- Built an Angular/FastAPI backend platform with REST APIs, async file processing, MongoDB workspace modeling, auth-aware flows, and AWS/Docker deployment paths.
- Designed typed REST contracts, service boundaries, repository/service layers, and background workers across UI and Business Service workflows.
- Implemented document ingestion/search services for upload, parsing, normalization, chunking, source tracking, OpenSearch, and MongoDB-backed retrieval.
- Added AI-enabled RAG/copilot workflows with scoped retrieval, citations, LLM orchestration, and production-safe answer grounding.

TECHNICAL SKILLS

Languages & Frameworks: Java 17, Python, Spring Boot, FastAPI, Pydantic, SQL, Scala, JavaScript/TypeScript

Backend & APIs: REST APIs, microservices, internal APIs, async workers, Java concurrency, service contracts, API integrations

Cloud & DevOps: AWS Lambda, Step Functions, ECS Fargate, S3, EC2, Docker, CloudFormation, Serverless, Jenkins CI/CD

Data & Streaming: Apache Flink, Kafka, Airflow, MemSQL, MongoDB, OpenSearch/Elasticsearch, batch/stream processing, indexing

Observability: Datadog, CloudWatch, Kibana, X-Ray, ELK Stack, data quality checks, alerting

AI-Enabled Systems: internal copilots, RAG, LLM integrations, vector search, scoped retrieval, citations, production support workflows

EDUCATION

State University of New York, New Paltz | Master of Science, Computer Science | CGPA: 3.6/4.0

May 2020

JNTUK | Bachelor of Technology | CGPA: 8.5/10

2018